



INFORMATION PAPER ON NERSA REGULATED JET FUEL INFRASTRUCTURE

Published: June 2026

Purpose of Publication

This Information Paper is published by the National Energy Regulator of South Africa (NERSA) to share high-level, non-confidential insights into South Africa's jet fuel infrastructure. It aims to promote transparency, support infrastructure planning, and enable informed engagement among stakeholders.

Disclaimer

This document is issued for information and knowledge-sharing purposes only and does not constitute a regulatory decision or policy position by NERSA.

Prepared by: Pheladi Makgamatha - Senior Engineer: Petroleum Pipelines Regulation
Contributor: Martin Untiedt - Head of Department: Petroleum Licensing Compliance & Dispute Resolution

Enquiries

Petroleum Pipelines Regulation Division
National Energy Regulator of South Africa (NERSA)
Email: liquidfuelstudy@nersa.org.za
Tel: +27 (0)12 401 4600
Website: www.nersa.org.za

Purpose of Publication

This Information Paper is published by the National Energy Regulator of South Africa (NERSA) to share high-level, non-confidential insights into the structure, utilisation, and emerging vulnerabilities of South Africa's jet fuel (Jet A1) infrastructure.

The intention of this publication is to:

- Promote transparency and knowledge-sharing within the liquid fuels sector;
- Provide evidence-based insights to support infrastructure planning and investment decisions;
- Enable informed engagement with industry participants, policymakers, and the public; and
- Contribute to the long-term security and resilience of aviation fuel supply in South Africa.

This paper forms part of Phase 1 of NERSA's broader Liquid Fuels Logistics and Infrastructure Vulnerability Study, with findings presented in a sanitised manner to protect commercially sensitive information while maximising public value.

Stakeholder Engagement and Enquiries

NERSA encourages stakeholders, industry participants, researchers, and interested parties to provide feedback or request further information related to this publication.

For enquiries or submissions, please contact:

Petroleum Pipelines Regulation Division

National Energy Regulator of South Africa (NERSA)

 **Email:** liquidfuelstudy@nersa.org.za

 **Tel:** +27 (0)12 401 4600

 **Website:** www.nersa.org.za

Reference: *Information Paper – Jet Fuel Infrastructure (June 2026)*

Call for Input

NERSA welcomes additional inputs, data contributions, and perspectives that may support ongoing phases of this study and enhance collective understanding of South Africa's liquid fuels infrastructure.

Information paper (non-binding, for knowledge-sharing purposes only)

TABLE OF CONTENTS

EXECUTIVE SUMMARY	4
PURPOSE AND SCOPE	5
Scope limitations:.....	5
REGULATORY CONTEXT	5
METHODOLOGICAL OVERVIEW	7
OVERVIEW OF AVIATION FUEL INFRASTRUCTURE (Sanitised).....	8
JET A1 THROUGHPUT PATTERNS AND TRENDS	12
LOGISTICAL FLOW OF JET A1 TO AVIATION FACILITIES	16
IDENTIFIED SYSTEMATIC VULNERABILITIES (high-level)	21
Pipeline Injections	21
Rail Injections	22
Road Transport.....	22
Regulatory matters.....	23
OPPORTUNITIES	24
ACSA Acquisition Strategy	24
Regulatory Insights and Opportunities	25
PRELIMINARY CONCLUSIONS AND RECOMMENDATIONS	25

LIST OF FIGURES

Figure 1: Representation of NERSA mandate within petroleum value chain	6
Figure 2: Small-sized Aviation Facilities	13
Figure 3: Medium-sized Aviation Facilities.....	14
Figure 4: Large-sized Aviation Facilities	15
Figure 5: ACSA Projected Sales Volumes.....	16
Figure 6: JetA1 product flow for road fed.....	17
Figure 7: Jet A1 Product Flow rail & pipeline	18
Figure 8: ORTIA Rail vs Pipeline Trend Volumes	19

LIST OF TABLES

Table 1: NERSA Licensed Storage Facilities licensed to handle Jet A1	9
Table 2: Licensed Aviation Facilities by NERSA	10

EXECUTIVE SUMMARY

South Africa's liquid fuels logistics system is undergoing structural change driven by refinery closures, divestment and ownership shifts in infrastructure, entry by independent wholesalers/distributors, and increasing reliance on refined product imports. In addition, global geopolitics have also impacted on South Africa in terms of product sourcing and security of supply, while the energy transition and decarbonization and move towards cleaner and more sustainable fuels is also factors playing a role. These changes introduce new dependencies and risks, particularly on aviation fuel (focus on Jet A1), as a niche market, which is characterised by stringent product quality requirements, limited supply points, and high-demand concentration at a small number of airports.

This information paper presents consolidated findings from Phase 1 of NERSA's *Liquid Fuels Logistics and Infrastructure Vulnerability Study*. The paper demonstrates how historical throughput and utilisation data submitted by licensed entities can be transformed into regulatory intelligence to support security of supply, infrastructure planning, and informed stakeholder engagement.

While Jet A1 is used as a case study, the analytical approach and regulatory insights are transferable across other liquid fuels and infrastructure segments regulated under the Petroleum Pipelines Act, 2003 (Act No. 60 of 2003) ('The Petroleum Pipelines Act').

PURPOSE AND SCOPE

The purpose of this paper is to:

- ❖ Share high-level, non-confidential insights derived from regulatory data;
- ❖ Illustrate a methodology for analysing licensed infrastructure utilisation;
- ❖ Highlight systemic infrastructure dependencies and vulnerabilities relevant to aviation fuel supply;
- ❖ Serve as a basis for infrastructure planning;
- ❖ Serve as a source of information for potential investors; and
- ❖ Support informed dialogue with policymakers, infrastructure operators, and other regulators.

Scope limitations:

- ❖ The paper does not assess commercial performance or pricing.
- ❖ Company-specific operational details have been aggregated or anonymised.
- ❖ The findings do not constitute regulatory determinations or compliance assessments.
- ❖ The overall project is ongoing, and therefore some of the information may also be subject to change and evolving.
- ❖ Data only considers NERSA licensed facilities.

REGULATORY CONTEXT

NERSA under the Petroleum Pipelines Act, 2003 (Act No 60 of 2003) ('the Petroleum Pipelines Act') regulates petroleum pipelines, storage facilities and marine loading facilities to promote efficient, secure, and equitable access to energy infrastructure.

In terms of section 1 of the Petroleum Pipelines Act, "*petroleum product means any liquid petroleum fuel and any lubricant, whether used or unused, and include any other substance which will be used for a purpose for which petroleum fuel or any lubricant may be used*".

In addition, a "*storage facility means any bulk storage facility and its auxiliary equipment that is or is intended to be used for the storage of petroleum and excludes storage facilities—*

- (a) *on the premises at which petroleum products are manufactured and where such storage facilities are integral to the process of manufacture;*

Information paper (non-binding, for knowledge-sharing purposes only)

- (b) for own final use;
- (c) used for the retailing of petroleum products to the public;
- (d) used by an agricultural cooperative exclusively for its members; and
- (e) used to transport petroleum by road, rail, sea and air.”

A representation of the portions of the value chain that NERSA regulates is shown below, highlighted within the orange selections.

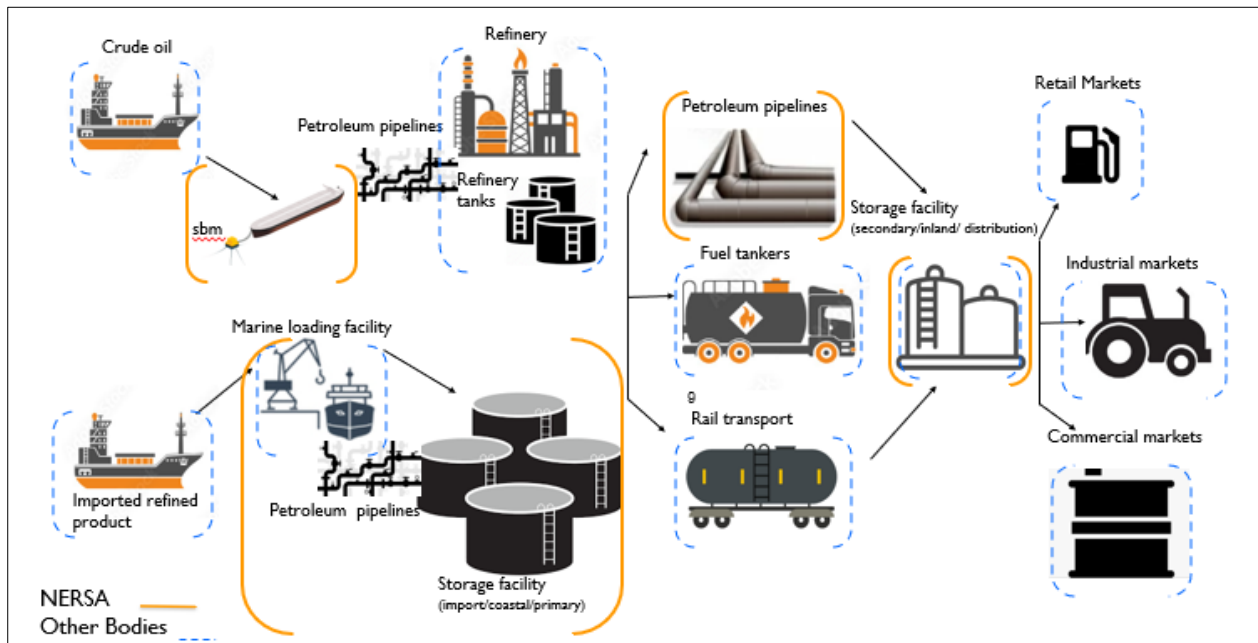


Figure 1: Representation of NERSA mandate within petroleum value chain

Therefore, it is against this backdrop that NERSA licences and regulates the storage of Jet A1 and infrastructure related to its transportation through pipelines and marine offloading facilities. As part of this mandate, the licence conditions and Regulations made in terms of the Petroleum Pipelines Act requires licensees to submit routine volume and utilisation data. Initially, such data has been used primarily for licensing and third-party access considerations. Given the reporting requirements, the data always lags and provides more of an historic picture.

This study repurposes that same data at a systems level to support proactive regulatory insight, moving beyond reactive decision-making towards evidence-based infrastructure planning.

METHODOLOGICAL OVERVIEW

Phase 1 focused on aviation fuel infrastructure serving South African airports. The analysis draws on:

- Historical throughput data reported by licensed storage and aviation facilities;
- NERSA Licensed capacity information; and
- Qualitative information obtained through regulatory engagements with infrastructure operators.

Facilities were grouped into small, medium, and large throughput categories to assess patterns of utilisation and logistical reliance. The emphasis is on trends over time, rather than individual facility performance.

OVERVIEW OF AVIATION FUEL INFRASTRUCTURE

There are eleven key points of entry or receipt for petroleum product into South Africa, categorised as follows:

- a) Inland Region:
 - o Mozambique via Matola;
 - o Namibia via Walvis Bay; and
 - o South African Refineries.
- b) Eastern Region:
 - o Port of Durban and
 - o Port of Richards Bay.
- c) Central Region:
 - o Port of Port Elizabeth (Gqeberha);
 - o Port of Ngqura; and
 - o Port of East London
- d) Western Region:
 - o Port of Mossel Bay;
 - o Port of Cape Town; and
 - o Port of Saldanha Bay.

In addition to the areas of entry, the Transnet Pipeline system, is the key transporter of liquid fuels from the Port of Durban to the inland region, as well as the two inland refineries to the inland area both for main fuels and Jet A1. The supply logistics linked to the supply points in terms of current operational local production (refining) facilities are as follows:

- Western Cape Province: Astron Energy (100 000bbl/day¹);
- Free State Province: NATREF Sasolburg (108 000bbl/day); and
- Mpumalanga Province: Sasol Synfuels (150 000bbl/day).

In terms of aviation fuel, South Africa's aviation fuel system is characterised by:

- A limited number of large airports accounting for the majority of national Jet A1 consumption;
- A network of smaller regional airports primarily supplied by road transport;

¹ The name plate capacities publicly available. Actual output may vary depending on operational requirements; and for Sasol the crude equivalent.

- Centralised coastal import and storage facilities acting as critical supply nodes; and
- Dependence on pipeline and rail for the inland supply to the current biggest airport, OR Tambo International Airport (ORTIA).

Most regional airports are supplied exclusively by road tankers, while the primary international airports rely on a combination of pipeline, road and rail infrastructure. Nationally, there is only a limited number of operational storage facilities that accommodate the handling of Jet A1. These facilities form the basis for receipt of aviation fuel from the operational refineries or from imports.

The list of the NERSA licensed facilities for the storage of Jet A1 is shown below.

Table 1: NERSA Licensed Storage Facilities licensed to handle Jet A1

Licensee	Region	Storage Type	Overall Capacity (m ³)	Approximate Jet A1 Capacity (m ³)	Mode of product intake
1. Engen	Island View B & C sites, Durban	Primary	262 970	46 611	Ship (imported [#])
2. bpSA and Shell jointly owned	Island View Site 1, Durban	Primary	81 441	16 946	Ship (imported)
3. Wasaa Terminals	East London	Primary	66 647	12 208	Ship (imported)
4. Burgan Cape Terminals	Cape Town	Primary	122 040	9 558	Ship (imported); Astron Refinery
5. bpSA and SFF JV	Milnerton	Secondary	86 684	16 873	Coastal shipping; Astron Refinery

Note[#]: can be locally (port to port) or internationally sourced

It is important to state that prior the Enref and SAPREF refinery closures, both the Engen and bpSA and Shell joint Island View facilities were fed through the refineries, supplemented by importation. The bpSA and SFF JV facility in Milnerton has the capability to store and receive JET A1 product from the bpSA and Shell facility in Island View through port-to-port product movement ('coastal imports'), albeit recent handling of JET A1 has been limited. In addition to these facilities, Engen discharges Jet A1 from both its Island View facilities and the Wentworth Enref tank farm and road gantry. The Enref tank farm tankage is not reflected in the above table due to its previous operations being done through its Manufacturing Licence prior to ceasing refining activities. As shown in earlier sections, storage tanks located on manufacturing premises and integral to manufacturing process, do not fall within the Petroleum Pipelines Act scope of regulation.

Therefore, storage capacity handling Jet A1 within the Astron refinery, previous Enref refinery and the inland NATREF refinery are not accounted for.

In addition to the primary and secondary storage facilities, NERSA had eleven (11)² airport storage facilities licensed and distributed across multiple provinces in South Africa. The table below represents the details of each airport facility with the current capacity as may be amended. Except for the Cape Town International Airport (CTIA) and OR Tambo International Airport (ORTIA), all aviation facilities store both aviation gasoline (avgas) and Jet A1, while the former two only store Jet A1. Additionally, ORTIA is the only airport that receives product through both rail and pipeline, while all the other airport facilities are road-fed.

Table 2: Licensed Aviation Facilities by NERSA

	Licensee	Airport	Location	Capacity (Design) [m ³]
1.	ACSA	OR Tambo International Airport (ORTIA)	Johannesburg, Gauteng Province	59 960
2.	ACSA	Cape Town International Airport (CTIA)	Cape Town, Western Cape Province	12 880
3.	ACSA	King Shaka International Airport (KSIA)	Durban, KwaZulu-Natal Province	6 000
4.	ACSA	Upington International Airport	Upington, Northern Cape Province	3 157
5.	ACSA	Bram Fischer Airport	Bloemfontein, Free State Province	494
6.	ACSA	George Airport	George, Western Cape Province	390
7.	ACSA	King Phalo Airport	East London, Eastern Cape Province	355
8.	ACSA	Kimberley Airport	Kimberley, Northern Cape Province	286
9.	bpSA ³	Rand Aviation Airport	Germiston, Gauteng Province	120
10.	bpSA	Cape Town Aviation	Cape Town, Western Cape Province	106
11.	bpSA	Plettenberg Bay Airport	Plettenberg Bay, Western Cape Province	60
Total Licensed Capacity				83 808

It is important to note that the list does not represent all the airports or aviation storage facilities within South Africa but rather shows those that are licensed by NERSA. More importantly, the Lanseria International Airport, which is a known significant consumer of Jet A1 is excluded due

² At the time of the study, some have been subsequently revoked as at May 2026

³ All the bpSA facilities (no 9,10 &11) have been revoked and are owned by new entities and due to their size and no longer licensed by NERSA

to it not being licensed by NERSA.

While several of previously licenced aviation facilities have revoked their licences for various reasons, many of the aviation facilities fall below the bulk determination storage threshold of 1500m³ required for NERSA licensing. On 26 May 2016, NERSA decided on what constitutes a 'bulk' storage facility as referred to in the Petroleum Pipelines Act. Under this determination, it was decided that facilities that are not connected to a pipeline or a marine loading facility and fall below a storage capacity of 1500m³ are eligible to be excluded from licensing. As such, majority of the aviation facilities are excluded from licensing, however, should an applicant require to be licensed, NERSA does consider its licensing. In this regard, Chief Dawid Stuurman International Airport (formerly Port Elizabeth Airport in Gqeberha) was previously determined to be a below bulk storage facility, but ACSA has since lodged an application with NERSA seeking an operation licence. Below is a presentation of the logistical map based on licensed NERSA facilities.

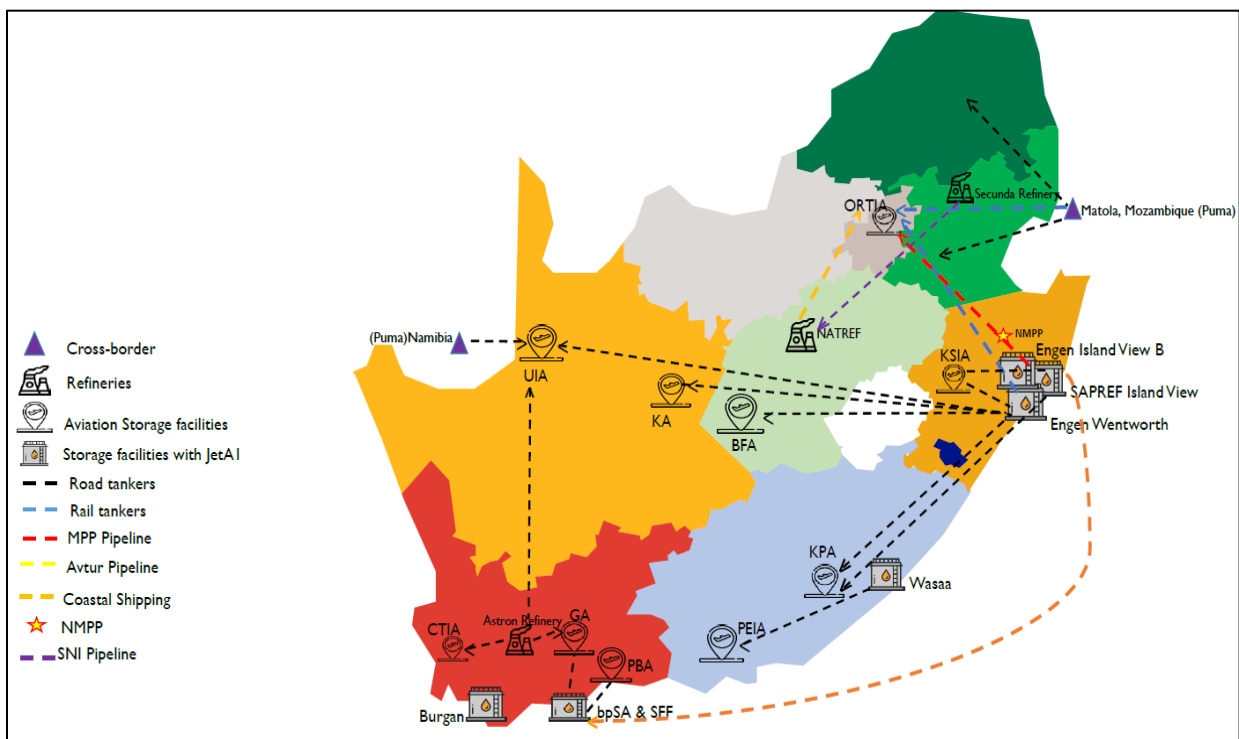


Figure 2: Jet A1 Logistics Map (NERSA adaptation)

JET A1 THROUGHPUT PATTERNS AND TRENDS

In this section, the focus is on the utilisation of the available storage facilities/capacity. Therefore, having due regard of the licensed aviation facilities, NERSA analysed the movement and throughput of Jet A1 from 1 April 2014 (2014/15 financial period FY) until 31 March 2023 (2022/23 FY) spanning over a period of 9 years. The analysis was based on availability of complete data sets.

From the analysis for the aviation facilities, a classification, into three categories was derived based on the facilities' annualized throughput volumes as follows: small, medium and large airports:

- a) Small < 10 000m³/annum.
- b) Medium 10 000 - 40 000m³/annum.
- c) Large > 40 000m³/annum.

The entire medium sized and small sized airports receive product via road/truck tankers. Only the ORTIA within the large airports receives product by both rail and pipeline, while CTIA and KSIA both receive through road tankers, similar to the small and medium aviation storage facilities. Six of the licensed aviation facilities were classified as small facilities based on their throughput of less than 10 000m³ annually. These facilities include the following:

- i. Bram Fischer Airport;
- ii. Kimberley Airport;
- iii. Rand Aviation Airport;
- iv. Upington International Airport;
- v. Cape Town Aviation; and
- vi. Plettenberg Bay Airport⁴.

The throughput trend of the above listed facilities is depicted in Figure 3 below. The actual throughputs are not shown for confidentiality and competition reasons.

⁴ BpSA Plettenberg Bay Airport – volume reports are only available from the 2021/22 period, hence this airport was excluded from the analysis in its entirety. This facility is in the process of being sold to CEM Air.

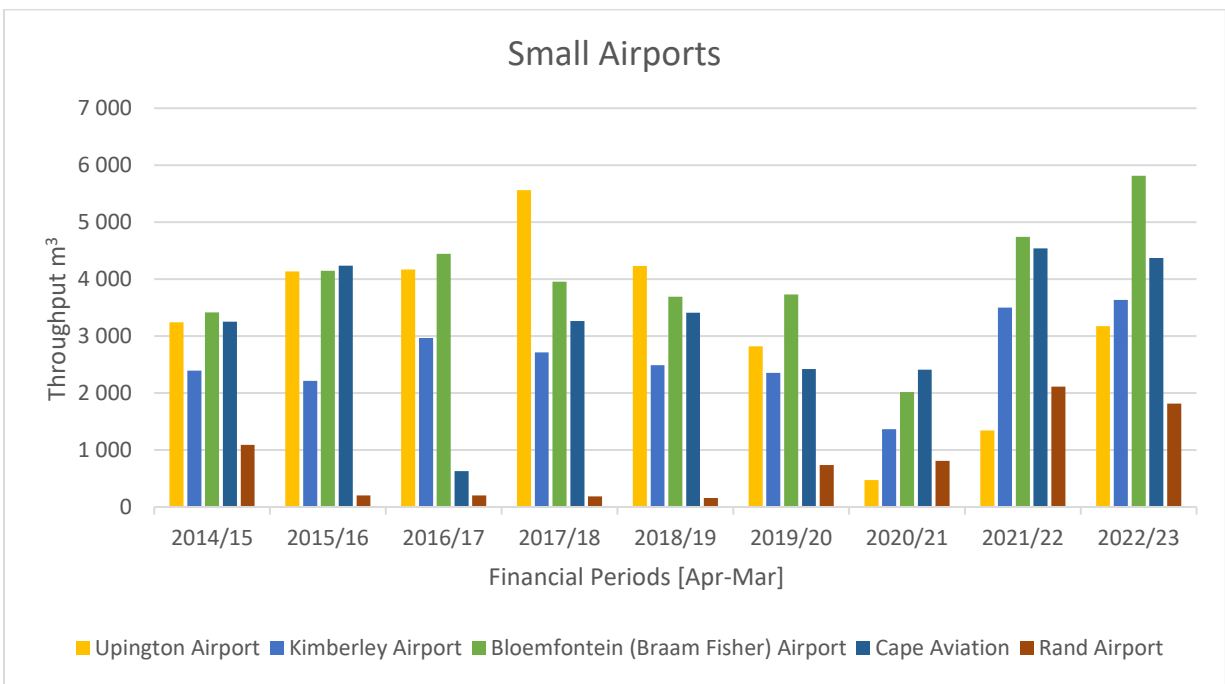


Figure 3: Small-sized Aviation Facilities

As can be seen in Figure 3, the majority of volume reports for the 2020/21 period are significantly lower in volumes in comparison to the other periods, due to the impact of COVID19 with travel restrictions nationwide. All the small-sized aviation storage facilities are road fed for product intake and on average throughput approximately 2 987m³/year based on the periods reviewed between 2014/15 until 2022/23. With the exception of the Uppington International Airport, all the other airports seem to be improving to levels even better than prior the COVID19 year/s.

Apart from the Cape Aviation and Rand Airport, which both house Air Schools, the other small aviation storage facilities are located on the 'outskirts' of the traditional routes.

Only two aviation facilities are classified within the medium-sized category, the East London (King Phalo) Airport and George Airport and both receive product via road tankers as well. Medium sized airports' throughput volumes amounting to between 10 000 - 40 000m³ per annum. The average throughput observed for these facilities is approximately 10 892m³ per annum, with facility turnovers on par with the NERSA benchmark of 2. This category of facilities does not seem to be constrained. Like small airports, the drop in FY2020/21 volumes was due to the COVID19

lockdown restrictions and has yet to recover to the pre-COVID19 levels.

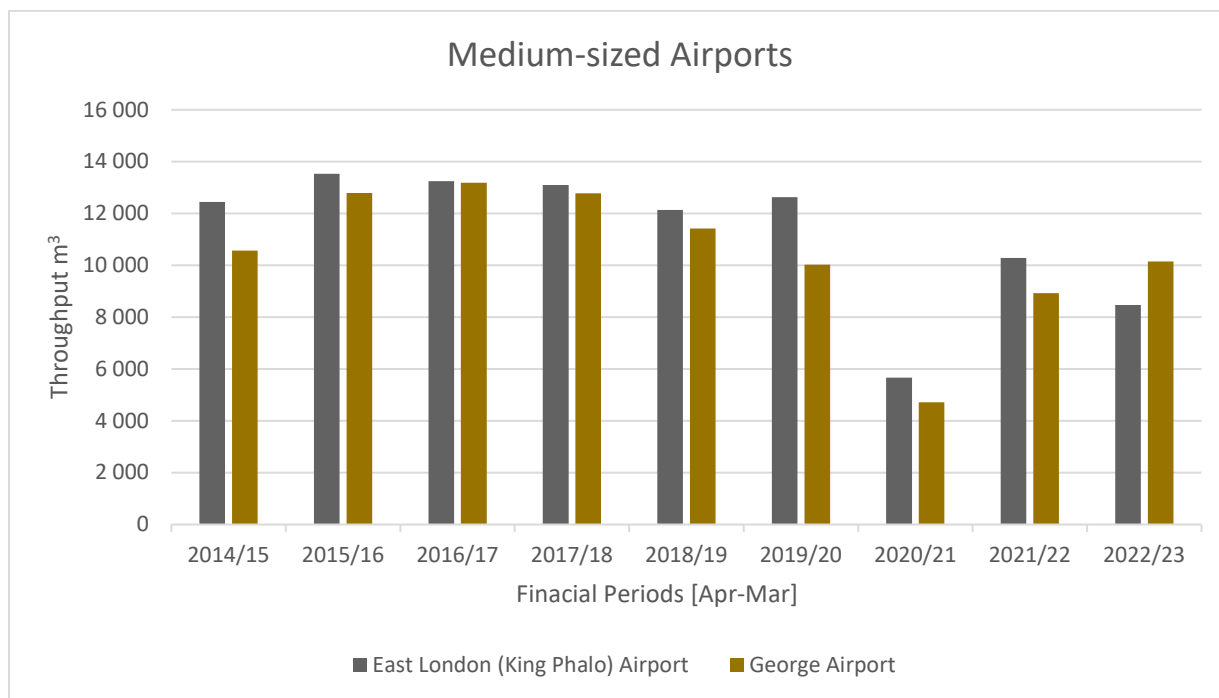


Figure 4: Medium-sized Aviation Facilities

The large airport category comprises of South Africa's key international airports, all having annual throughputs above 40 000m³ per annum. However, even with all three distinctly large, the difference in the quantum of throughput is significant, and as such an annualized average or median for all three combined would substantially skew the throughput. Therefore, the sum of the actual annual throughput was utilized to determine an average of 1 871 200m³ per annum. Relative to the entire annualised average of throughput for all sized airports, the large airports make up 98% of the entire throughput.

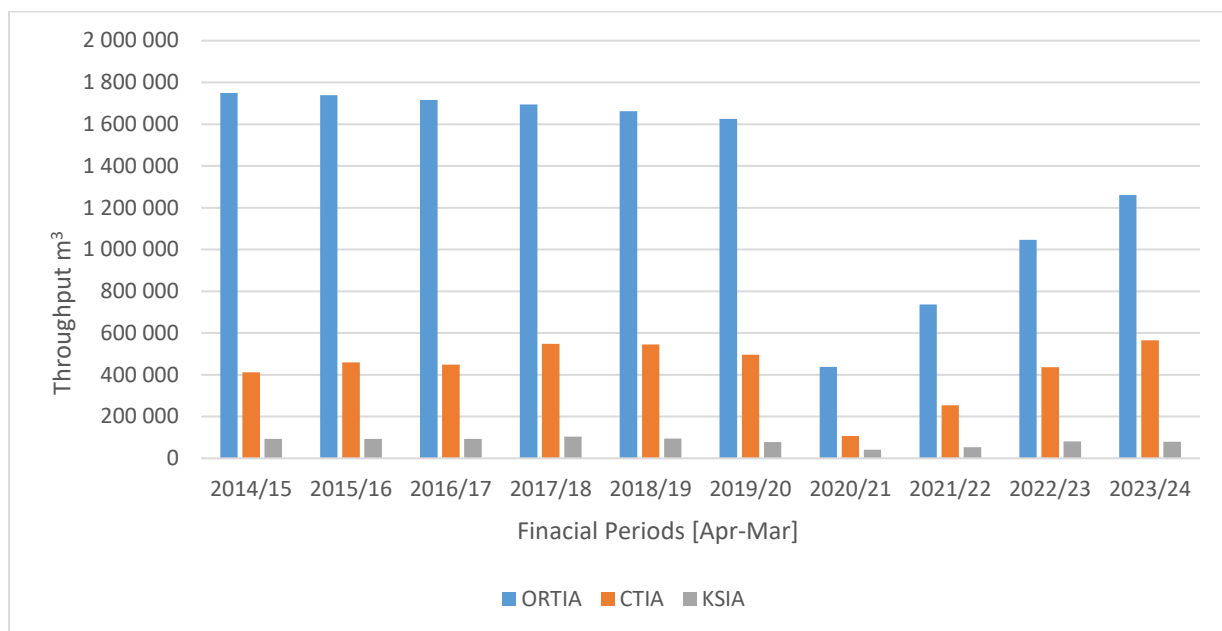


Figure 5: Large-sized Aviation Facilities

As can be seen from Figure 5 above, in terms of consumption, the aviation facilities did not quite get back to pre-COVID levels of throughput by the end of the 2023/24 period, except for the CTIA, which had surpassed pre-COVID levels. According to ACSA, full recovery across the board is expected in the 2025/26 financial period and thereafter consistent steady growth annually is expected based on the Fuel Master Plan. The projected sales volumes up to 2032 by ACSA are shown below.

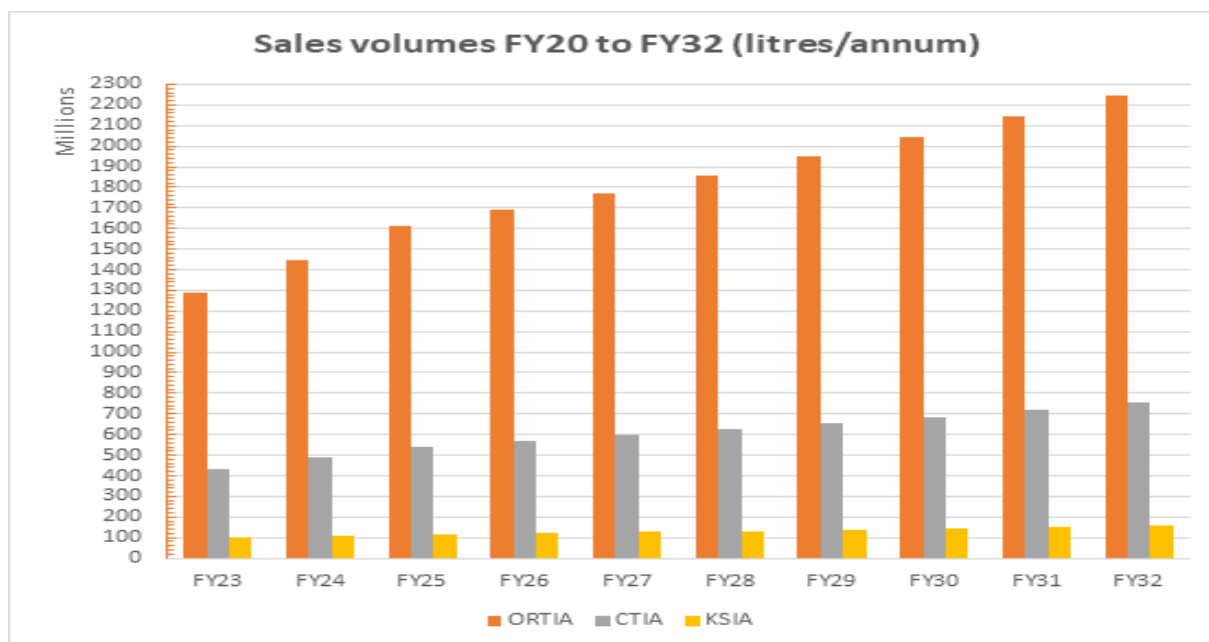


Figure 6: ACSA Projected Sales Volumes

The key aggregate observations based on the above classifications include the following:

- Large aviation facilities account for the overwhelming majority of national Jet A1 throughput;
- Medium and small facilities exhibit relatively stable demand patterns but are more exposed to road logistics disruptions;
- System-wide demand experienced a temporary decline during the COVID-19 period, followed by gradual recovery;
- Recovery trajectories vary by airport size and connectivity to major logistics corridors.

These patterns highlight the importance of differentiated regulatory attention based on infrastructure role rather than nominal capacity alone.

LOGISTICAL FLOW OF JET A1 TO AVIATION FACILITIES

Based on the information available to NERSA, the standard logistical flow of aviation fuel in South Africa is such that product is sourced either directly from the manufacturers and refineries or it is imported through the ports and points of entry and trucked to the respective aviation facilities. A simple diagram representing the product flow for Jet A1 is shown below.

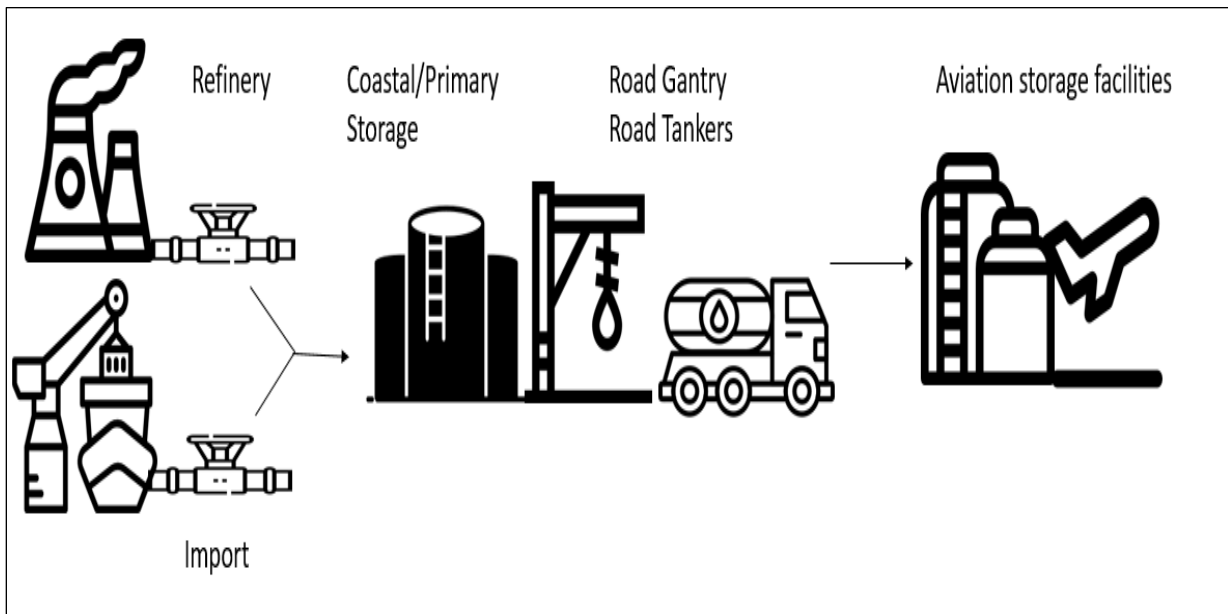


Figure 7: JetA1 product flow for road fed

Apart from ORTIA, all other aviation facilities receive product in this manner⁵. What can be seen from this representation is that the aviation facilities play both the part of secondary storage and final destination of the product consumption, unlike the typical secondary storage facilities that still store product for further re-distribution. Generally, there has not been significant product flow logistical challenges for aviation facilities that receive product through the road tankers.

However, at CTIA in Cape Town under peak demand conditions/and abnormal operational circumstances/shutdowns, the timeous replenishment via truck ex-Astron Energy refinery has been borderline (marginal) which resulted in volumes dropping below the agreed stock holding levels. Jet A1 can also be supplied to the CTIA storage facility through the Burgan Cape Terminals (BCT) storage facility, which can take direct imports.

In addition, the Kimberley, Upington and Braam Fisher airports all receive product hauled by road tankers from the Engen Wentworth and Island View facilities, the Astron Energy refinery and from Walvis Bay in Namibia.

⁵ **Gqeberha** (Port Elizabeth) previously also received product via pipeline from Dom Pedro storage Complex to the airport, but the line was decommissioned as far back as 2008 also considering that Dom Pedro was supposed to have closed down many years ago. **ORTIA** is fed in three different pipelines (refer to paragraph 6.12).

On the other hand, product receipt for ORTIA through rail and pipeline is depicted in Figure 8 below.

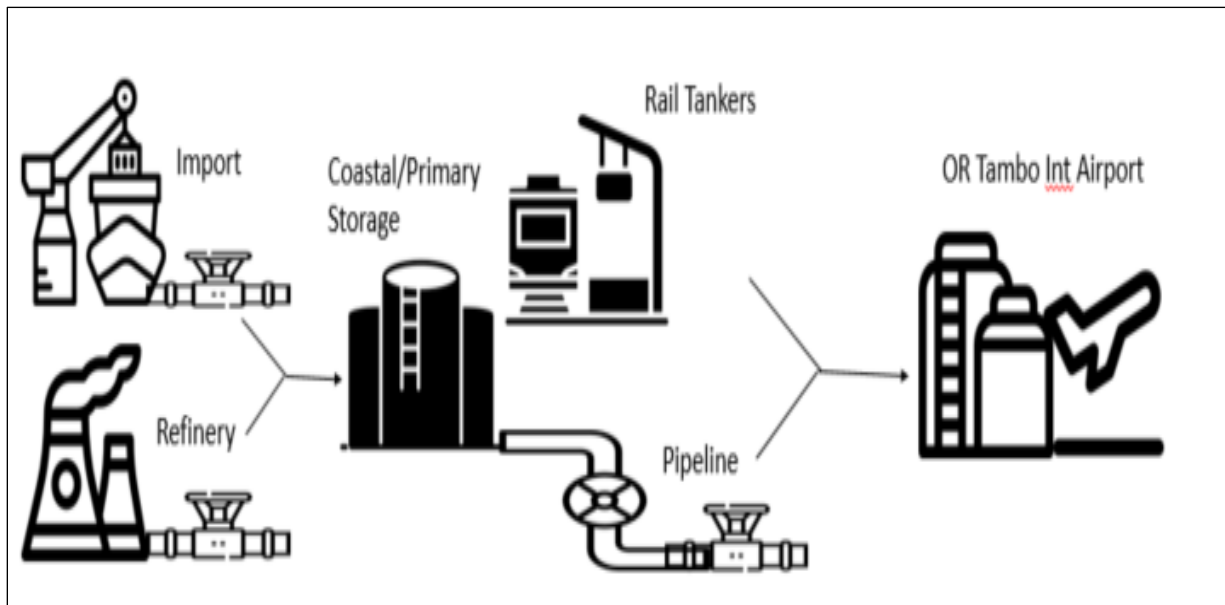


Figure 8: Jet A1 Product Flow rail & pipeline

ORTIA, the largest SA airport, receives product via the both the Multi-Product Pipeline (MPP) from the Durban coast via Jameson Park and through the Aviation Fuel Pipeline (*Avtur* pipeline) directly from the Natref Refinery in Coalbrook, passing through Meyerton, to ORTIA; as well as through rail (via TFR Isando). Amongst the large airports, ORTIA is the only facility that utilizes both pipeline and rail for product receipt. Below are figures that depict the split between rail product movement and pipeline product movement for each year of the specified periods. What can be observed from the two figures, is the gradual move away from rail receipt (2015/16) with the trend over the period moving from 30:70 to roughly 10:90. This should be viewed with the context of the use of block trains to supply ORTIA during the period when the old Transnet Durban Johannesburg pipeline was the constraint (circa 2019)

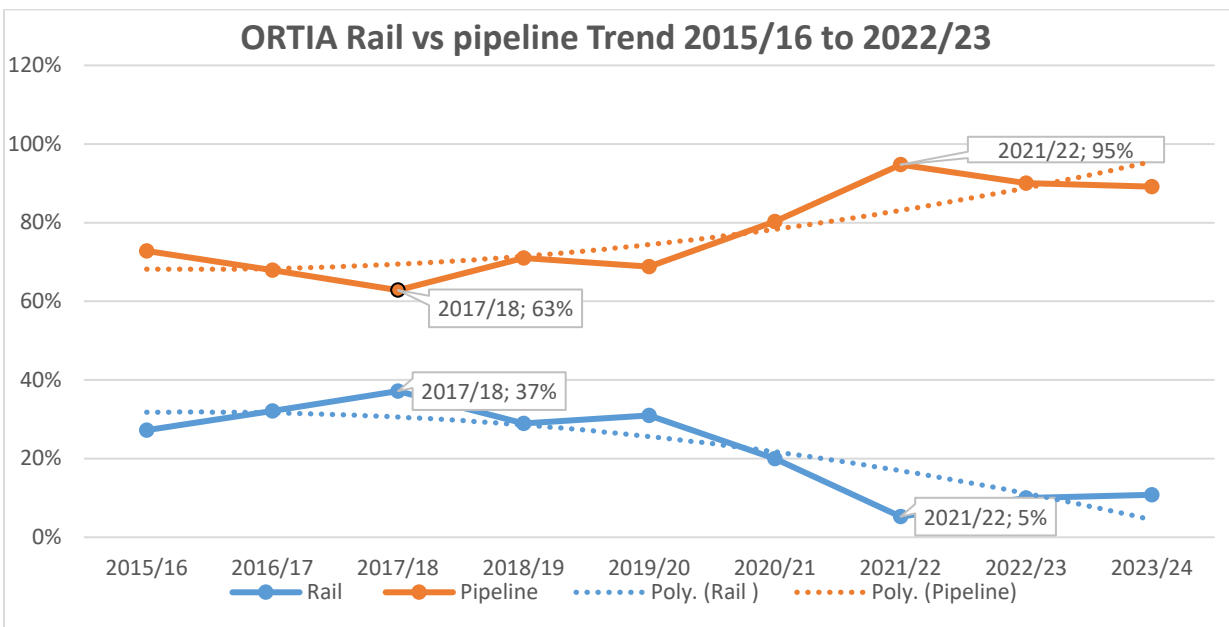


Figure 9: ORTIA Rail vs Pipeline Trend Volumes

According to ACSA, it does not foresee an increment back to higher utilisation for rail mode of transport into the future. However, discussions regarding future projects on both rail and road receipt for ORTIA are currently taking place, since the required volumes are only feasible with either rail and/or pipeline as mode of supply. According to industry participants, within the rail recorded volumes, roughly 60-69% of rail product is received through the Matola corridor, with the remaining balance from Durban.

From Durban, product is transported via rail from the storage facilities in Island View [Engen and SAPREF both have rail gantries]. During the period of the DJP constraints, this was the preferred option with block trains running to the Transnet Freight Rail (TFR) rail siding in Isando, near ORTIA to supply the ORTIA facility. The rail offloading gantries are linked to the pipeline inlet manifold owned by the jointly owned ORTAFS Consortium⁶ (bpSA, Engen, Excel, Shell & Total); and the 1 890m long pipeline runs underground crossing the freeway and emerges within the ORTIA storage premises. The turnaround time for this type of rail product receipt from Durban is estimated to take 168 hours (7-8 days), according to TFR.

⁶ The consortium operate the storage and provide refueling services to airlines on a lease basis. Approximately 90% of the consumption is supplied in this manner (ORTAFS stands for O.R. Tambo Airport Fuelling Services).

Various industry participants have made requests for road offloading gantries at ORTIA to alleviate the rail delays which they have stated as having lead times of on average 33 days to deliver product to the airport while the service design has a maximum 14 days. However, as indicated in earlier paragraphs, rail volumes are not at this stage anticipated to get back to their historic numbers and the proposal to have road receipt via trucks for ORTIA is considered at this preliminary stage to not be cost effective given the consumption of Jet A1 at ORTIA of at least 3 million litres a day. Nevertheless, ACSA has indicated that it is investigating various product flow options.

In addition to rail, ORTIA is supplied through pipelines in two other ways:

Option 1 is from NATREF with the dedicated 6-inch Aviation turbine fuel ('Avtur') pipeline in the northern network from Coalbrook passing through Alrode towards the ORTIA. Product from the NATREF refinery can also be sourced from the Sasol Oil Secunda refinery (Synthetic Jet A1) and then fuel components transferred through the Secunda to Natref (SNI) pipeline which is linked to the NATREF site.

Important to note that theoretically any third party (wholesalers supplying the airlines at ORTIA) can transport product via the Transnet pipeline network based on the *common carrier* principle. However, practically and operationally, injection of Jet A1 into the Avtur line is only possible from storage tanks (finished goods) within the Natref refinery. As a result, the product needs to be procured from either Sasol or Total Energies ex-Natref refinery or in terms of product swaps agreements;

Option 2 is through the Transnet MPP southern section via Jameson Park to Alrode (new MPP section) through the remaining 12-inch section of the DJP pipeline between Alrode and ORTIA (this portion of the pipeline is not yet decommissioned but is nearing the end of its useful asset life and will then need to be replaced by MPP PL-6 section). Transnet has ceased operations of the remaining sections of the ORTIA pipeline with effect 31 December 2023 and its decommissioning will commence in due course. In addition to this, products can be pumped from Jameson Park to Coalbrook NATREF using the 18-inch line (DJP) where product is then transferred to the 6-inch dedicated NATREF line to ORTIA (semi-3rd option). This option includes some operational challenges including volume losses because of product interface

(intermixture) and the handling/ treatment of this.

Therefore, in terms of the logistics related to aviation fuel, the analysis identified three primary modes of Jet A1 supply as follows:

- a) **Pipeline supply** – efficient for high-volume demand but dependent on limited injection points and strict quality controls due to the pipeline handling both aviation and ground fuels (diesel and petrol) and availability of ground fuels to push out the injected Jet A-1;
- b) **Rail transport** – historically significant but increasingly constrained by operational reliability challenges;
- c) **Road transport** – flexible but not feasible as a primary supply mode for very high demand volumes.

The concentration of demand and reliance on a small number of supply corridors introduce systemic vulnerabilities, particularly during extreme events or infrastructure outages.

IDENTIFIED SYSTEMATIC VULNERABILITIES (high-level)

Pipeline Injections

In terms of Jet A1 injection through the MPP, product is fed from only two facilities, and specific tanks that are dedicated Special Storage Warehouse (SOS) tanks licensed by SARS under the Customs and Excise Act. The tanks are specifically licensed for the storage of imported goods for the operation of a duty and tax-free warehouse, and hence the specification and dedicated tanks. Therefore, Jet A1 injection from the coast can only be done through those specific facilities.

In addition, there is currently insufficient storage capacity at the ports to accommodate Jet A1, particularly when supply to ORTIA is routed through the MPP during periods when the Natref refinery is offline and all volumes earmarked for ORTIA are imported via the Port of Durban.

Furthermore, for every jet fuel product transfer into the MPP, one can lose up to 1 million litres worth of product in the front and at the back (can be re-blended for other fuels). This is a conservative treatment to ensure product quality and safety based on the Joint Inspection Group (JIG) standard for aviation fuel quality management. According to industry, it takes approximately 2 days to check and verify product quality, the first is done at the point of entry in Durban and the

next at Jameson Park before product can be transferred to the ORTIA. It takes approximately another 4 days to cross pump from Jameson Park to ORTIA. Therefore, the delays herein have a potential to negatively affect swift receipt of product through the MPP.

Over and above the potential delays, Jet A1 handling requires a delay in handling to allow the product to first settle in specified tanks before the same product can be transferred to other tanks for further consumption.

Rail Injections

The main challenge noted with product receipt through rail tank cars is the significant delays of product movement due to the integrity of the rail infrastructure. Furthermore, the rail tank cars are controlled by different parties based on the areas of jurisdiction, even though the same infrastructure is used. In the metros, the Passenger Rail Agency of South Africa (PRASA) is responsible for the control on the scheduling of the rail tank cars, while TFR mainly oversees the long-haul commodity scheduling. However, due to the reliance of the same rail line infrastructure, sequencing can cause delays.

TFR has also suffered from theft and vandalism of its rail infrastructure, resulting in a lack of reliance of its service. Unfortunately, NERSA does not have oversight on rail infrastructure and details therein.

Road Transport

Although majority of the aviation storage facilities receive product via road trucks, road bridging is not really feasible to meet large demand for the large airports and thus limited and mainly used during supply disruptions (e.g. refinery and/or pipeline challenges). The safety risks associated with using road transport are very high and results in high numbers of truck on the South African roads.

Regulatory matters

There are different role players and authorities involved in the supply of aviation fuels and a coordinated approach is necessary to address some of the infrastructure vulnerabilities. This includes the need for keeping strategic stocks for jet fuel which has been highlighted during incidents of supply stockouts arising from the Durban floods and refinery shutdowns (planned or unplanned) (DMPR/CEF). The issues pertaining to the Customs & Excise Act also needs to be addressed with SARS. Investment in additional infrastructure to import, store and transport Jet A1 is needed by the industry and providing certainty of regulatory matters may assist to unlock that. A more pro-active infrastructure planning approach is needed vs responding to applications on a case-by-case basis.

Without reference to specific operators or assets, the identified vulnerabilities highlight the following:

- Dependency on imported product;
- The small number of coastal injection points and storage capacity for aviation fuels for inland supply;
- Limited substitutability between logistics modes at short notice;
- Exposure of high-demand airports to refinery, import facility, pipeline and rail disruptions;
- Regulatory fragmentation across fuel quality, customs, transport, and infrastructure oversight.
- The importation of dual-purpose kerosene ['DPK'] (C11-C14), giving oil companies the flexibility to channel product into aviation and/or domestic fuels uses depending on pricing, bringing further uncertainty to continuous jet fuel supply.

These vulnerabilities are structural rather than operator-specific and require coordinated, cross-institutional responses/interventions.

OPPORTUNITIES

ACSA Acquisition Strategy

One of the key developments in the aviation industry is ACSA's Acquisition of Regional Fuel Infrastructure and Equipment Strategy to acquire regional airports where it historically held long term land lease agreements with the oil companies. The long-term leases enabled the oil companies to establish fuel infrastructure to service the jet fuel and avgas requirements at the respective airports. However, with the lease agreements nearing their end of term, ACSA decided to exercise its rights to acquire the fuel storage facilities and related assets at these airports.

ACSA stated that the objective of its acquisition strategy is to:

- a) Provide transparency of the jet fuel processing costs with plans to consult industry to improve efficiency;
- b) Apply for NERSA fuel infrastructure ownership and operation licences to ensure continuity of fuel supply at ACSA's Regional Airports; and
- c) Ensure that ACSA complies with its mandate of supplying fuel to all aircrafts that may use its airports.

Through various engagements with NERSA, ACSA indicated that some of the drivers for the implementation of the acquisition strategy is due to the logistical challenges for the storage and supply of jet fuel in recent times. ACSA indicated that with no capital investment programs being made to expand the South African refining capacity, airport growth could be restricted and as such has decided to promote the ease of supply of jet fuel to airlines through airport infrastructure acquisitions.

ACSA further explained that with the overlap of the hydrocarbon range between jet fuel (C9-C14) and diesel (C11-C16), in-country refiners could opt to not sell jet fuel to parties that supply Jet A1 to ACSA's airports when diesel prices are more attractive or price jet fuel beyond that of the jet fuel global pricing structure.

It is important to note, however, that since acquiring the various strategic storage facilities, ACSA has initiated fuel masterplan studies aimed at identifying bottlenecks within fuel infrastructure across its fuel farms, particularly in relation to storage, at all nine airports it owns and operates. The insights and recommendations from these studies are being used to develop a structured

programme to refurbish and expand storage capacity and associated infrastructure where necessary, ensuring alignment with projected demand growth.

Regulatory Insights and Opportunities

The Phase 1 findings demonstrate that:

- Regulatory data can be leveraged to identify system-level risks early;
- Infrastructure utilisation metrics provide valuable signals for future investment needs;
- Proactive information sharing among regulators and policymakers can reduce supply risk;
- Geographic and flow mapping tools can enhance regulatory oversight and planning.

The study also underscores the importance of transitioning from case-by-case infrastructure approvals towards more integrated, corridor-based planning approaches.

PRELIMINARY CONCLUSIONS AND RECOMMENDATIONS

Phase 1 of NERSA's logistics study illustrates the value of transforming routine regulatory data into strategic infrastructure insight. By sharing these findings in a controlled and sanitised manner, NERSA contributes to improved transparency, informed stakeholder engagement, and enhanced resilience of South Africa's liquid fuels supply system.

Future work will expand this approach across additional regions and infrastructure segments to further support evidence-based energy regulation. Although this phase focused on aviation fuel, the methodology is applicable to:

- Coastal import terminals;
- Inland storage depots;
- Multi-product pipeline systems; and
- Cross-border fuel corridors.

Subsequent phases of the broader study will apply similar analytical techniques to other regions, corridors and fuel categories.

END